

## **COURTNEY M. CURRIER, Ph.D.**

### ***Curriculum Vitae***

Postdoctoral Research Associate, University of Cambridge  
Department of Plant Sciences, Downing Street, Cambridge CB1 3EA United Kingdom  
[cc2228@cam.ac.uk](mailto:cc2228@cam.ac.uk) | <https://courtcurrier.github.io>

## **RESEARCH INTERESTS**

I am an ecosystem ecologist at the University of Cambridge focusing on carbon storage potential and persistence in grasslands and savannas. I am interested in understanding how climate change and increasing disturbances, such as fire, are affecting cycling of essential nutrients between plants and soils in dryland ecosystems, and how this underlies changes to carbon dynamics. Tools I use are fire and rainfall manipulation experiments in the field, synthesis of spatial datasets, and stable and radioisotopes.

## **ACADEMIC QUALIFICATIONS**

2018–2023     Ph.D. Environmental Life Sciences, Arizona State University, Tempe, AZ  
2013–2015     M.Sc. Biology, Arizona State University, Tempe, AZ  
2009–2013     B.Sc. Environmental Sciences (Honors Research), University of Notre Dame, Notre Dame, IN

## **PROFESSIONAL EXPERIENCE**

Jan 2026       Postdoctoral Research Associate, Earth System Science, Stanford University  
Jul 2023–  
present       Postdoctoral Research Associate, Department of Plant Sciences, University of Cambridge, United Kingdom  
Jan 2025–  
Mar 2025     Lecturer, University of Cambridge  
Aug 2017–  
Aug 2018     Project Manager, Global Drylands Center, School of Sustainability, Arizona State University  
Dec 2015–  
Aug 2018     Research Specialist and Lab Manager, Sala Laboratory, Arizona State University

## **FELLOWSHIPS AND GRANTS**

(\$62,290 earned)

2023           [rejected] National Science Foundation Postdoctoral Fellowship in Biology: Bridging the gap between ecosystem vulnerability to nitrogen loss and opportunity for uptake  
2022           [rejected, semifinalist] Presidential Management Fellowship  
2022           PhD Completion Fellowship, School of Life Sciences, Arizona State University  
2022           Graduate College Travel Award, Arizona State University

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|-----------|----------------------------------------------------------------------------------------------------------------|
| 2020      | Student Registration Award, Ecological Society of America Biogeosciences Section                               |
| 2019–2021 | Jornada LTER Graduate Research Fellowship                                                                      |
| 2019      | Student Travel Award, Ecological Society of America Southwest Section                                          |
| 2019      | Research Training Initiative Grad Student Support Grant, Arizona State University                              |
| 2018–2019 | Arizona State University Graduate College Fellowship                                                           |
| 2018      | Excellence in Rangeland Ecology Research Travel Award, Ecological Society of America Rangeland Ecology Section |
| 2017      | Travel Grant Award, ASU, Graduate and Professional Student Association                                         |
| 2015      | Travel Grant Award, ASU, School of Life Sciences                                                               |
| 2015      | Travel Grant Award, ASU, Graduate and Professional Student Association                                         |
| 2013      | [rejected, Honorable Mention] National Science Foundation – Graduate Research Fellowship                       |
| 2012      | National Science Foundation – Research Experience for Undergraduates Fellowship, Niwot Ridge LTER              |
| 2012      | Travel Grant Award, University of Notre Dame, College of Science                                               |
| 2012      | Travel Grant Award, National Science Foundation – Biology REU                                                  |

## HONORS AND AWARDS

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|-----------|---------------------------------------------------------------------------------------------------------------------------------|
| 2022      | [semifinalist] Achievement Rewards for College Scientists                                                                       |
| 2021      | “Grad Slam” Elevator Pitch 3 <sup>rd</sup> Place Winner, Arizona State University Graduate and Professional Student Association |
| 2020      | Outstanding Research Award, Arizona State University Graduate and Professional Student Association                              |
| 2018–2022 | The Graduate College Excellence Award, Arizona State University                                                                 |
| 2013      | Outstanding Environmental Scientist Award, University of Notre Dame                                                             |
| 2012–2013 | Honors Research Student in Biological Sciences, University of Notre Dame                                                        |
| 2011–2012 | Dean’s List, University of Notre Dame, College of Science                                                                       |
| 2011      | Clare Boothe Luce Undergraduate Research Award, University of Notre Dame College of Science Summer Research Fellow              |

## PEER-REVIEWED PUBLICATIONS

(H index = 8 | [Google Scholar](#))

**Currier, C.M.**, Parrinello, C., Dick, C., Augustine, D.J., Blair, J.M., Boughton, E.H., Coetsee, C., David, A., Derner, J.D., Hobbie, S.E., Isbell, F., Mkansi, N., Modimola, K.S., Reich, P.B., Wigley, B. Worm, K., Zeglin, L.H., and A.F.A. Pellegrini. 2025. Climate versus consumer controls on above- and belowground productivity in savannas. In preparation for submission to *Nature Ecology & Evolution*. Full draft available upon request.

- Currier, C.M.**, Reichmann, L.G., and O.E. Sala. 2025. Unresponsive drylands to nitrogen availability: Access to alternative sources? In preparation for submission to *Journal of Geophysical Research: Biogeosciences*. Full draft available upon request.
- Currier, C.M.**, Reichmann, L.G., and O.E. Sala. 2025. Acclimation of the nitrogen cycle to changes in precipitation. In Review (minor revision). *Ecosystems*. Manuscript available upon request. [DOI](#)
- Biancari, L., ..., **Currier, C.M.**, ..., et al. 2024.. Drivers of woody dominance across global drylands. *Science Advances* 10,eadn6007. [DOI](#)
- Díaz-Martínez, P., ..., **Currier, C.M.**, ..., et al. 2024. Vulnerability of mineral-associated soil organic carbon to climate across global drylands. *Nature Climate Change* 14: 976–982. [DOI](#)
- Maestre, F., ...**Currier, C.M.**, ..., et. al. 2022. Grazing and ecosystem service delivery in global drylands. *Science* 378: 915-920. [DOI](#)
- Currier, C.M.** and O.E. Sala. 2022. Precipitation versus temperature as phenology controls in drylands. *Ecology*. [DOI](#)
- Osborne, B. B., Bestelmeyer, B. T., **Currier, C. M.**, Homyak, P. M., Throop, H. L., Young, K., and Reed, S. C. 2022. The consequences of climate change for dryland biogeochemistry. *New Phytologist*. [DOI](#)
- Currier, C.M.**, Makings, E., Anderson, J., Slagel, K., and J. Maranville. 2021. Solanaceae Part Seven: Browallia, Calibrachoa, Capsicum, Jaltomata, and Salpichroa. *Canotia* 17: 46-60. [DOI](#)
- Currier, C.M.**, Chaloner, D.T., Ruegg, J., Tiegs, S.D., D’Amore, D., and G.A. Lamberti. 2020. Beyond macronutrient resource subsidies: Pacific salmon (*Oncorhynchus* spp.) as potential vectors of micronutrients. *Aquatic Sciences* 82:50. [DOI](#)
- Sala, O.E., Boone, C.G., Turner II, B.L., and **C.M. Currier**. 2019. The sustainability gap and its implications. *Current Opinion in Environmental Sustainability* 39: 39-43. [DOI](#)
- Delgado-Baquerizo, M., ..., **Currier, C.M.**, ..., et al. 2019. Changes in belowground biodiversity during ecosystem development. *Proceedings of the National Academy of Science* 116(14): 6891-6896. [DOI](#)
- Currier, C.M.**, and J.J. Elser. 2017. Beyond monoculture stoichiometry studies: assessing growth, respiration, and feeding responses of three *Daphnia* species to P-enriched, low C:P lake seston. *Inland Waters* 7(3): 348-357. [DOI](#)
- Rüegg, J., **C.M. Currier**, D.T. Chaloner, S.D. Tiegs, and G.A. Lamberti. 2014. Habitat influences Pacific salmon (*Oncorhynchus* spp.) tissue decomposition in riparian and stream ecosystems. *Aquatic Sciences* 76(4): 623-632. [DOI](#)

## SELECTED CONFERENCE PRESENTATIONS

(last five years)

**Currier, C.M.**, Parrinello, C., Dick, C., Augustine, D.J., Blair, J.M., Boughton, E.H., Coetsee, C., David, A., Derner, J.D., Hobbie, S.E., Isbell, F., Mkansi, N., Modimola, K.S., Reich, P.B., Wigley, B. Worm, K., Zeglin, L.H., and A.F.A. Pellegrini. 2025. Fire and aridity controls on soil carbon inputs. Baltimore, MD, USA. August 2025. (talk)

**C.M. Currier** and O.E. Sala. Nitrogen plant uptake in a semiarid ecosystem is modulated by water and plant type. Ecological Society of America Annual Meeting. Montreal, Quebec, Canada. August 2022. (talk)

**C.M. Currier** and O.E. Sala. A temporal perspective of nitrogen cycling in a semiarid grassland under extreme rainfall conditions. Ecological Society of America Annual Meeting. Virtual. August 2021. (talk)

**C.M. Currier** and O.E. Sala. Re-envisioning plant phenology: From data points to painting. Ecological Society of America Annual Meeting. Virtual. August 2020. (Inspire talk, co-organizer of session)

**C.M. Currier** and O.E. Sala. Extreme precipitation interacts with N cycling in a semi-arid grassland. Ecological Society of America Annual Meeting. Virtual. August 2020. (talk)

## INVITED TALKS

(last five years)

Archbold Biological Station, Microbial Ecology Workshop. August 2025.

Emmanuel College, University of Cambridge, College Research Symposium. February 2025.

Department of Plant Sciences, University of Cambridge. February 2025.

University of the Third Age, Botany Seminar. January 2025.

Cambridge Botanical Garden, Science on Sunday Series. July 2024.

## SYNERGISTIC ACTIVITIES

2024–present College Research Associate, Emmanuel College, University of Cambridge

2024–present Women in Soil Ecology Mentor

2022 Abstract Reviewer for the Ecological Society of America Southwestern Chapter

2020–2022 Grant Reviewer for the Graduate and Professional Student Association, Arizona State University

2019–2023 Vision Kids Art Program Instructor, Chandler, AZ

2019–2021 Jornada Basin LTER Graduate Student Representative

2019 IsoCamp, University of Utah

2018 Explore National Ecological Observatory Network Workshop, Boulder, CO

2017–present BIODESERT Global Research Group

2017–present Alumni Mentor, Environmental Sciences undergraduate program, Notre Dame

2016–present Asombro Institute, New Mexico State University, Jornada LTER, volunteer

2013–2014 Arizona State University, School of Life Sciences Graduate Executive Committee, secretary

2013–present Arizona State University Ask a Biologist, volunteer

2012–2013 Notre Dame Biological Sciences Senior Leadership Committee

## OTHER PUBLICATIONS

**C.M. Currier** and T. Feltus. 2023. Finding joy through art, ecology, and bikes. Feature in *The Radavist*. [Online only](#).

**C.M. Currier**. 2019. Feature in *Phoenix Transect*. [Online only](#)

**C.M. Currier** and O.E. Sala. 2018. Art provides a new lens for moving through the scientific method. *SciArt Magazine* 33: How can art influence science? [Online only](#)

**C.M. Currier**. 2018. Feature in *Life Raft Zine*. [Online only](#)

## GROUP ART EXHIBITIONS

Paintings: <https://courtecurrier.github.io/fieldwork.html>

*THIRST 2022: Artists for Humanitarian Aid*. [Online only](#). July 2022 (original paintings)

*Dimensions of Science*. Onyx Gallery, Phoenix, AZ. February 2019 (original painting)

*Portraits of Science*. MonOrchid Gallery. Phoenix, AZ. May 2017 (original paintings)

## TEACHING

*University of Cambridge Lecturer and Supervisor*

Part II Plant Sciences Module: Global Change

*Graduate Teaching Assistant at Arizona State University*

(\*indicates significant role in course development)

Biology 426: Limnology\*

Biology 320: Fundamentals of Ecology\*

Biology 281: General Biology I for Majors

Biology 282: General Biology II for Majors\*

Biology 181: General Biology I

Biology 182: General Biology II

Biology 100: The Living World

*Undergraduate Teaching Assistant at Notre Dame*

Biology 30312: General Ecology